WORKSHEET

Chapter 9: Practical applications

1 John runs a landscaping supplies business. For an order of mulch, he charges $60 for delivery, plus $25 per cubic metre of mulch.

a Complete this table of values for the relationship.

No. of m^3 of mulch delivered, M	1	2	3	4	5
Total cost of order, C ($)	85	110			

b What is the independent variable in this relationship?

c Graph this data on a Cartesian plane.

d What is the vertical intercept of this graph?

e Find the linear relationship between M and C.

f Use your answer from part **e** to calculate the total cost of a delivery of:

i 7 m^3 of mulch.

ii 25 m^3 of mulch.

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2 A conveyor belt carries objects from the bottom of a mine shaft to the Earth's surface, rising at a rate of 5 m/s. The mine shaft is 150 m deep.

a Complete this table of values

Time, *t* (seconds)	0	1	2	5	10
Depth below the Earth's surface, *d* (metres)					

b What is the independent variable in this relationship?

c Graph the data in the table on a Cartesian plane.

d Could this graph be extended beyond $t = 10$s?

e What would be the significance of the horizontal intercept?

f Find the equation of the linear relationship between t and d.

g Use your answer from part **f** to find how long it takes for an object placed on the conveyor belt at the bottom of the shaft to reach the surface.

3 Micah started up a clothing store. Here initial costs were \$46 000, and her ongoing costs per month are \$15 300.

a What is the independent variable in this relationship?

b Complete this table of values

Time, *t* (months)	0	1	2	3	4
Total cost, *C* (\$)					

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c Graph the data in the table on a Cartesian plane.

d Find the linear relationship between t and C.

e What will be the cost of running Micah's store for a year?

4 The following equation shows the relationship between temperature in degrees Celsius and temperature in degrees Fahrenheit:

$$F = 32 + \frac{9}{5}C$$

a Use the equation above to complete this table of values (giving values to one decimal place).

Temperature (°C)	−10	0	10	20	30	40
Temperature (°F)						

b Use the equation to convert 100 °C to degrees Fahrenheit.

c Graph this data on a Cartesian plane, extending the horizontal axis to 100 °C, and the graph to this point.

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d Use the graph to answer the following:

i What is 80 °C equivalent to in degrees Fahrenheit?

ii What is 140 °F equivalent to in degrees Celsius?

5 Adele invests $1500 in the bank at a simple interest rate of 9% p.a. for 20 years. The relationship between the value of Adele's investment, A (in $) and time, t (in years) is given by the equation $A = 1500 + 135t$.

a Complete this table of values using the information above:

Time, t (years)	0	1	2	3	10	15
Value of Investment, A ($)						

b What is the independent variable in this relationship?

c Use the data in the table to construct a graph showing the value of Adele's investment over 20 years.

d Use the equation to determine the final value of the investment at the end of 20 years.

6 A taxi driver charges $3.50 plus $2.15/km for any trip.

a What is the independent variable in this relationship?

b Use the information above to complete this table of values:

Distance travelled, d (km)	0	1	2	3	10	15	50
Cost, c ($)	3.50	5.65					

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c Graph the data in the table on a Cartesian plane.

d Use the graph to answer the following questions:

i How much will a trip of 20 km cost?

ii Marcus has $30 to spend on a taxi fare. How many kilometres can he afford to travel in one trip?

e Determine the equation of the linear relationship between d and c.

7 One Australian Dollar (AUD) is equivalent to 0.77 euros (EUR)

a Complete the following table

Australian Dollars (AUD)	1	2	5	10	20
Euros (EUR)					

b What is the dependent variable in this relationship?

c Graph the data in the table on a Cartesian plane.

d Determine the equation of the linear relationship between Australian Dollars and euros. To avoid confusion, use D for dollars and E for euros.

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e Use your answer from part **d** to determine:

i the value of AUD50 in euros

ii the value of AUD276 in euros

iii the value of EUR70 in Australian dollars

iv the value of EUR5000 in Australian dollars.

8 Susie is a plumber. She charges $60 for arriving at a job, and $75 per hour that she works.

a What is the independent variable in this relationship?

b Use the information above to complete this table:

Time worked, *t* (hours)	0	1	2	3	10
Cost, *C* ($)					

c Graph the data in the table.

d What is the significance of the intercept of the vertical axis?

e Determine the equation of the linear relationship between t and C.

f Use your graph to answer the following:

i How much would Susie charge for working 7 hours?

ii How long has Susie worked if she charges $360?

g Use your answer to part **e** to determine how much Susie would charge for working on a job for a total of 12 hours?

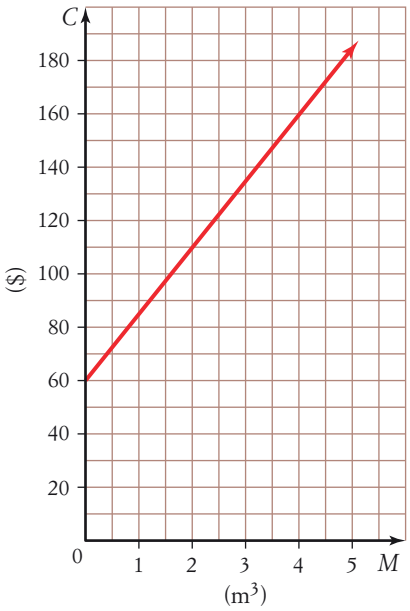
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Answers

1 a 135, 160, 185

b M

c



d 60

e $C = 60 + 25M$

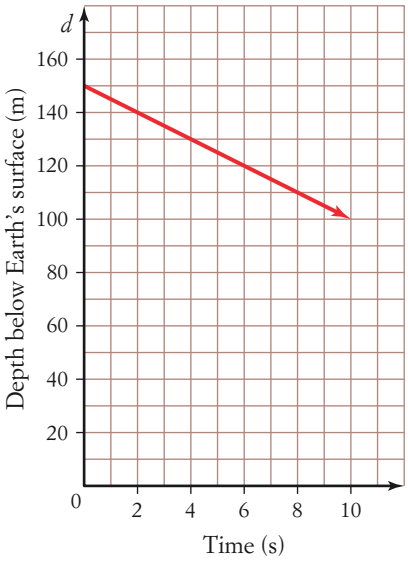
f i \$235

ii \$685

2 a 150, 145, 140, 125, 100

b d

c



d Yes

e This would tell us how long it would take for an object to reach the surface from the bottom of the mine.

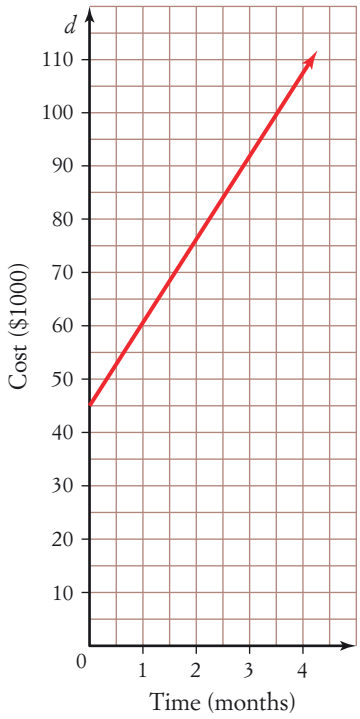
f $d = 150 - 5t$

g 30 seconds

3 a Cost

b 46 000, 61 300, 76 600, 91 900, 107 200

c



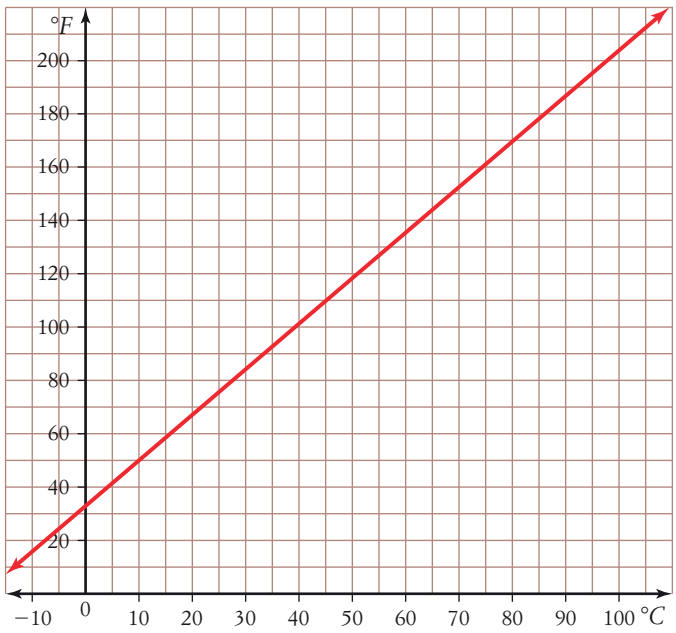
d $C = 46\,000 + 15\,300t$

e 229 600

4 a 14, 32, 50, 68, 86, 104

b 212 °F

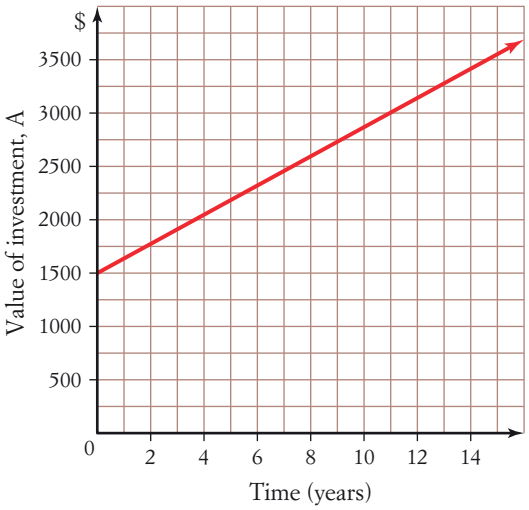
c



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d i ≈170 °F
ii 62 °C approx

5 a 1500, 1635, 1770, 1905, 2850, 3525
b Time
c



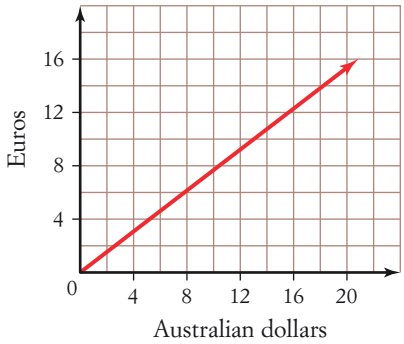
d $4200

6 a distance
b 7.80, 9.95, 25, 35.75, 111
c



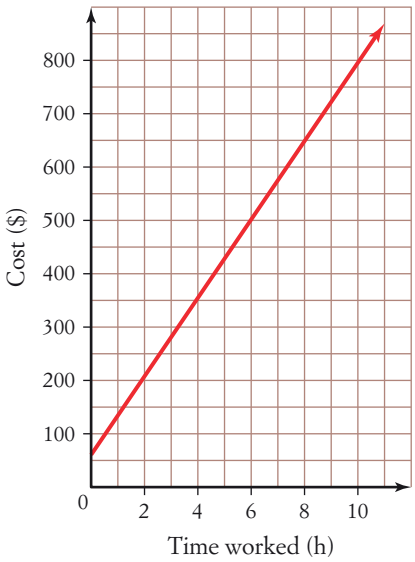
d i ≈ $46.50
ii ≈ 12 km
e $c = 3.5 + 2.15d$

7 a 0.77, 1.54, 3.85 7.70, 15.40
b euros
c



d $E = 0.77D$
e i €38.50
ii €212.52
iii A$90.91
iv A$6493.51

8 a time
b 60, 135, 210, 285, 810
c



d This is what Susie charges for arriving at a job.
e $C = 60 + 75t$
f i $585
ii 4 hours
g $960